

- 3 (a) By considering the binomial expansion of $(z+z^{-1})^4$, where $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that $\cos^4 \theta = \frac{1}{8}(\cos 4\theta + 4 \cos 2\theta + 3)$. [5]

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- (b) Use the substitution $x = \sin \theta$ to find the exact value of $\int_0^{\frac{1}{2}} (1-x^2)^{\frac{3}{2}} dx$. [3]

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4 The integral I_n is defined by $I_n = \int_0^1 (1+x^5)^n dx$.

(a) By considering $\frac{d}{dx}(x(1+x^5)^n)$, or otherwise, show that

$$(5n+1)I_n = 2^n + 5nI_{n-1}. \quad [5]$$

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.

(b) Find the exact value of I_3 .

[4]

[illegible]

- 7 (a) Use the substitution $u = x^2 - 1$ to find $\int \frac{x}{\sqrt{x^2 - 1}} dx$. [3]

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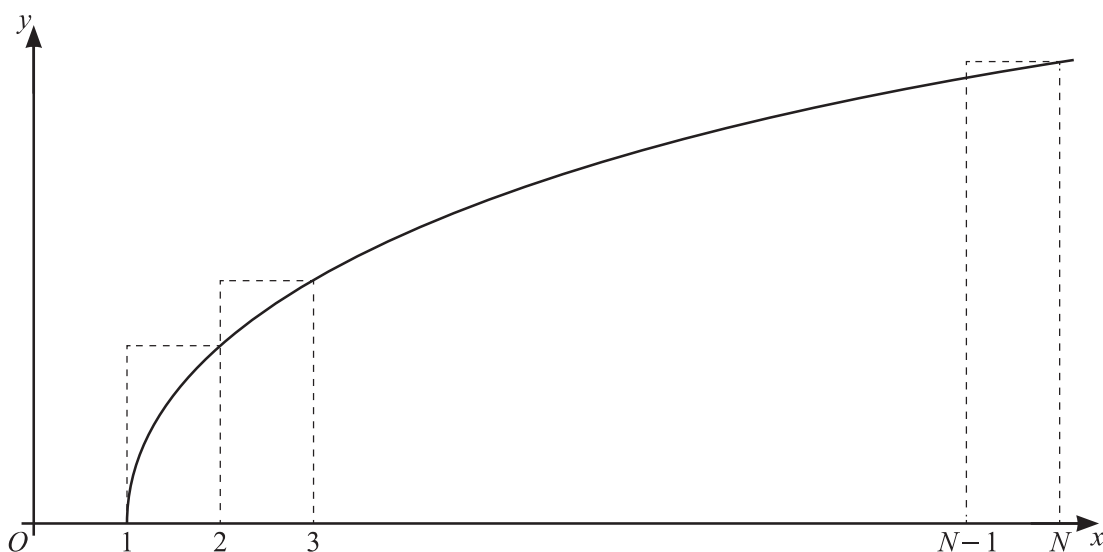
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The diagram shows the curve with equation $y = \cosh^{-1} x$ together with a set of $(N-1)$ rectangles of unit width.

- (b) By considering the sum of the areas of these rectangles, show that

$$\sum_{r=2}^N \ln(r + \sqrt{r^2 - 1}) > N \ln(N + \sqrt{N^2 - 1}) - \sqrt{N^2 - 1}. \quad [5]$$

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- (c) Use a similar method to find, in terms of N , an upper bound for $\sum_{r=2}^N \ln(r + \sqrt{r^2 - 1})$. [3]

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